Experiment-5(logistics delivery network)

Aim:

To find the least-cost delivery route between two nodes using the Uniform Cost Search algorithm.

Algorithm:

1. Initialize a priority queue with the start node and cost = 0.
2. While the queue is not empty:
    - Pop the node with the lowest cost.
    - If it is the goal, return the path and total cost.
    - Otherwise, expand its neighbors and calculate their cumulative costs.
    - Push neighbors into the priority queue if not visited.
3. If the queue empties without reaching the goal, report failure.

Python Program:

```
python
import heapq

graph = {
   'A': [('B', 2), ('C', 5)],
   'B': [('C', 1), ('D', 4)],
   'C': [('D', 1)],
   'D': []
}

def uniform_cost_search(start, goal):
   pq = [(0, start, [start])]
   visited = set()

   while pq:
      cost, node, path = heapq.heappop(pq)

      if node == goal:
         print("Delivery route:", path)
         print("Total cost:", cost)
         return

      if node in visited:
         continue

      visited.add(node)

      for neighbor, weight in graph[node]:
```

```
            if neighbor not in visited:
                heapq.heappush(
                    pq,
                    (cost + weight, neighbor,
                     path + [neighbor])
                )

    print("Route not found")

uniform_cost_search('A', 'D')
```

Sample Output:

Code
Delivery route: ['A', 'B', 'C', 'D']
Total cost: 4